PCA on Stock Returns and Economic Interpretation

Group 9

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1 Introduction

During major market events, individual stock returns tend to move together in ways that suggest a lower-Dimensional structure underlying the data. The COVID-19 crash and the 2022 interest-rate cycle are prominent examples: broad macroeconomic shocks propagated across sectors simultaneously, producing return co-movement that cannot be explained solely by firm-specific factors. This pattern motivates a central question in empirical asset pricing: whether a small number of latent common factors can account for the shared variation in equity returns.

Financial economics offers a well-established theoretical framework for identifying those factors. The Fama-French three-factor and five-factor models propose that equity returns are systematically explained by exposure to the market portfolio, a size premium, a value premium, and, in the five factor extension, profitability and investment. These factors are understood to represent pervasive sources of risk that affect broad cross-sections of stocks.

This project investigates the empirical generalizability of the Fama-French factor framework asking whether its theoretical risk structure is latent in price data itself. Given nothing more than a time series of returns for 20 large-cap U.S. stocks from 2016 to 2025, we apply Principal Component Analysis to extract latent factors from the return correlation matrix without imposing any prior economic structure, and ask whether the components that emerge correspond to the risk dimensions Fama-French theory predicts.

2 Objectives

This project pursues three analytical objectives. First, we apply PCA to standardized daily log-returns for 20 large-cap U.S. stocks to identify the dominant latent factors driving shared return variation, assessing how many components are needed to capture the bulk of the covariance structure. Second, we interpret the leading components economically through stock loadings and regression analysis, and validate these interpretations against the Fama-French three- and five-factor models. Third, we evaluate whether the factor structure translates into superior portfolio performance by constructing mean-variance tangency portfolios and comparing them out-of-sample against SPY and an equal-weight benchmark, and examine whether the factor structure itself remained stable across the 2016–2022 and 2023–2025 subperiods, a period marked by NVIDIA's emergence as a dominant return driver.

3 Data and Methodology

Let $P_{i,t}$ denote the adjusted closing price of stock i on day t. Daily log-returns are computed as $r_{i,t} = \log(P_{i,t}/P_{i,t-1})$. We form a return matrix R where each column corresponds to one stock. To remove differences in volatility scale, returns are standardized prior to PCA:

$$z_{i,t} = \frac{r_{i,t} - \bar{r}_i}{s_i}, \tag{1}$$

where $\bar{r}_i$ and s_i are the sample mean and standard deviation of stock i. PCA is applied to the correlation matrix of standardized returns. Under the eigen-decomposition $\Sigma = V\Lambda V^\top$, the columns of V are the principal component loading vectors, Λ contains the eigenvalues, and the explained variance ratio of component k is $\lambda_k / \sum_{j=1}^{p} \lambda_j$, where $p = 20$.

To interpret the extracted factors, we examine stock loadings and regress individual returns on the PCA scores. For Apple: $r_{\mathrm{AAPL},t} = \beta_0 + \beta_1 PC_{1t} + \beta_2 PC_{2t} + \varepsilon_t$. We then assess whether the PCA factors recover known risk dimensions by regressing each component on the Fama-French factors. For FF3:

$$PC_{k,t} = \alpha_k + b_{1k}(Mkt\text{-}RF)_t + b_{2k}SMB_t + b_{3k}HML_t + \varepsilon_{k,t}, \tag{2}$$

and for FF5, RMW_t and CMA_t are added:

$$PC_{k,t} = \alpha_k + b_{1k}(Mkt\text{-}RF)_t + b_{2k}SMB_t + b_{3k}HML_t + b_{4k}RMW_t + b_{5k}CMA_t + \varepsilon_{k,t}. \tag{3}$$

A high R^2 indicates that PCA, despite operating without prior economic labels, recovers factors resembling established sources of systematic risk.

4 Results

The introduction argued that broad macroeconomic shocks produce return co-movement inconsistent with 20 independent series, and that this co-movement may reflect a small number of latent common factors. The data bear this out directly.

Cumulative returns over the sample period are highly dispersed across stocks. The distribution is strongly right-skewed, driven primarily by NVDA and TSLA, and exhibits high kurtosis, consistent with fat-tailed return behavior that makes normality-based risk models unreliable. We therefore use historical Value-at-Risk throughout. The equal-weighted portfolio experienced a maximum drawdown of −38.41%, concentrated during the COVID-19 crash (Appendix).

Figure 1 shows that this drawdown did not occur in isolation; average pairwise correlation across the 20-stock universe spiked above 0.75 during the COVID-19 crash, and remained elevated through the 2022 interest-rate cycle. Outside of these episodes, correlation falls back toward 0.2–0.3. That a diversified 20-stock portfolio lost over a third of its value in a single episode reflects not idiosyncratic risk but pervasive co-movement: stocks across sectors fell together because they share common exposures. The full pairwise structure is documented in (Appendix), where within-sector clustering among financials ($\rho > 0.80$) and technology is particularly pronounced. Together these patterns confirm that 20 return series are not independent; a small number of latent common factors likely account for the bulk of shared variation, which we now extract via PCA.

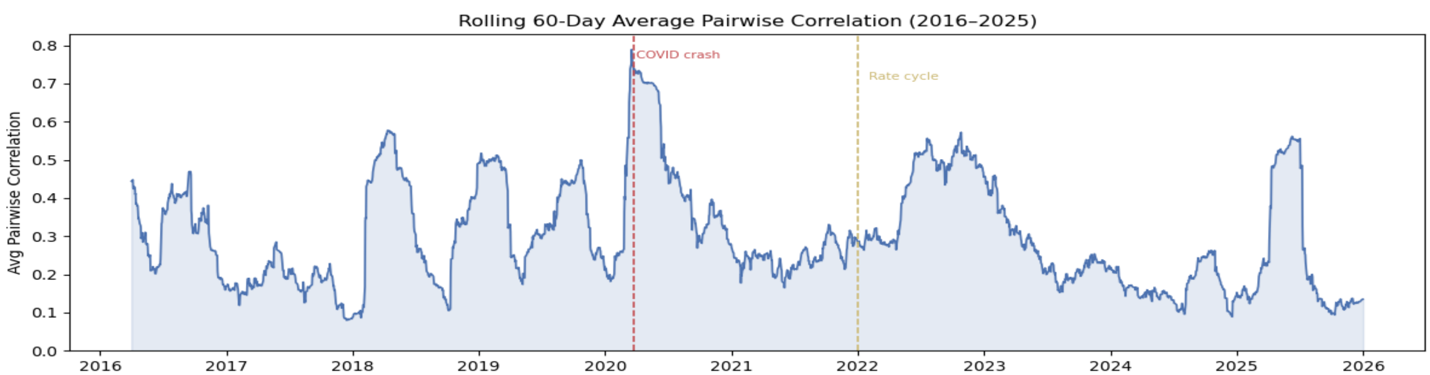

Figure 1: Rolling 60-day average pairwise correlation across the 20-stock universe, 2016–2025. Correlation spikes sharply during the COVID-19 crash (March 2020) and remains elevated through the 2022 interest-rate cycle, consistent with common macroeconomic shocks driving broad return co-movement. Outside of stress episodes, average correlation reverts toward 0.2–0.3.

4.1 PCA Results

We apply PCA to the correlation matrix of standardized daily log-returns for the 20-stock universe. The explained variance drops sharply after the first component and flattens after the fifth (Appendix), indicating that a small number of factors drives most of the shared return variation. The first five principal components explain approximately 70.4% of total variance, with individual contributions of 41.9%, 12.8%, 7.8%, 4.2%, and 3.7% for PC1 through PC5 respectively. The dominance of PC1 alone suggests a single pervasive source of common risk, consistent with a broad market factor affecting all stocks simultaneously.

Figure 2 shows the loadings of the first three components for each stock, sorted and colored by sector. PC1 loads positively across all 20 stocks with no exceptions, which is the defining signature of a market-wide factor; every stock in the universe moves in the same direction along this dimension. PC2 exhibits a clean sign reversal:

growth and technology stocks (AMZN, NFLX, META, NVDA, MSFT, GOOGL, AAPL) load positively while financials and energy stocks (XOM, CVX, WFC, JPM, BAC) load negatively, consistent with a growth-versus-value style factor that captures sector rotation. PC3 tells a different story: consumer staples and healthcare stocks (PEP, KO, JNJ) load strongly positively while most other sectors load near zero or negatively, suggesting this component captures a defensive quality dimension related to profitability and investment stability.

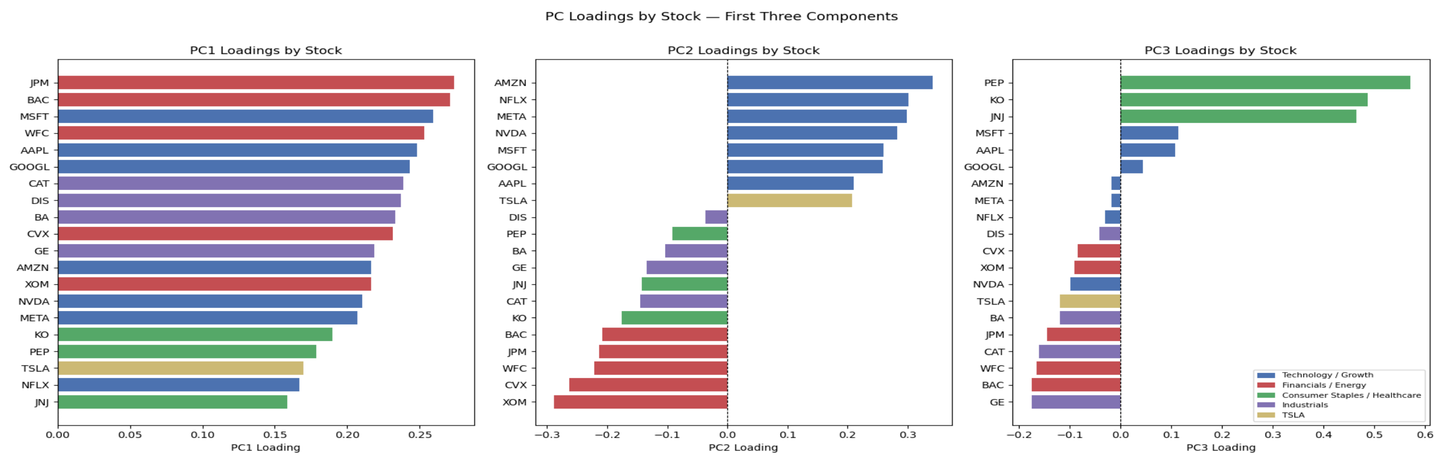

Figure 2: Loadings of the first three principal components, sorted and colored by sector. PC1 loads uniformly positive across all stocks. PC2 separates growth and technology from financials and energy. PC3 is dominated by consumer staples and healthcare, suggesting a defensive quality dimension.

4.2 Regression Using PCA Factors

To test whether the leading PCA components explain individual stock returns, we use Apple as a case study, regressing daily returns on PC1 and PC2:

$$r_{\text{AAPL},t} = \beta_0 + \beta_1 PC_{1t} + \beta_2 PC_{2t} + \varepsilon_t.$$

The estimated model yields an R^2 of 0.629, meaning the two data-driven factors account for 62.9% of Apple's daily return variation. Both coefficients are highly statistically significant: PC1 has a t-statistic of 59.1 and PC2 of 27.7, confirming that Apple's returns are jointly driven by broad market risk and growth-style exposure. The F-statistic of 2,127 ($p \approx 0$) rejects the null that neither factor matters. Notably, both PC1 and PC2 enter with positive coefficients, consistent with Apple's classification as a large-cap technology stock that co-moves strongly with the market and loads on the growth side of the value-growth dimension. These results demonstrate that PCA factors carry meaningful explanatory power at the individual stock level, not only in summarizing the full return matrix.

4.3 Fama-French Validation

We validate the PCA interpretation by regressing each of the first five components on the Fama-French three- and five-factor models. Table 1 below summarizes the adjusted R^2 under both specifications and the improvement from FF3 to FF5.

Table 1: Adjusted R^2 from Fama-French Regressions on Each Principal Component

	Adjusted R^2		
PC	FF3	FF5	Δ
PC1	0.946	0.948	+0.002
PC2	0.650	0.675	+0.025
PC3	0.145	0.226	+0.081
PC4	0.005	0.036	+0.032
PC5	0.021	0.028	+0.007

PC1 is almost entirely explained by the market factor under both specifications ($R^2 = 0.946$ under FF3, 0.948 under FF5), with a market factor t-statistic of 205.6, by far the largest in the model. The FF5 extension adds virtually nothing, confirming that PC1 is a pure market factor. Its correlation with SPY (0.960), the equal-weight portfolio (0.991), and the Fama-French market factor (0.958) further corroborates this interpretation.

PC2 is primarily explained by HML, the value-minus-growth factor, with a coefficient of -139.6 ($t = -67.4$) under FF3. The negative sign is consistent with the loading pattern: growth stocks load positively on PC2 while value stocks load negatively, so PC2 moves opposite to HML. Under FF5, R^2 rises modestly from 0.650 to 0.675, with RMW and CMA adding incremental power. PC2's raw correlation with HML of -0.804 makes the value-growth interpretation unambiguous.

PC3 shows the largest improvement from FF3 to FF5, with adjusted R^2 rising from 0.145 to 0.226. Both RMW ($t = 14.1$) and CMA ($t = 10.0$) are highly significant under FF5, suggesting that PC3 captures a profitability and investment dimension absent from FF3, consistent with the loading pattern, where defensive consumer staples and healthcare stocks dominate its positive end. PC4 and PC5 have low explanatory power under both models ($R^2 < 0.04$) and likely reflect residual sector-specific or idiosyncratic variation rather than systematic risk.

Taken together, the Fama-French validation confirms the interpretations suggested by the loading patterns: PC1 corresponds to market risk, PC2 to value-growth style rotation, and PC3 to a profitability and investment dimension. PCA, operating without any prior economic labels, recovers the same factor structure that decades of asset pricing research have identified as the dominant sources of systematic equity risk.

Supporting figures including the FF3 and FF5 factor loadings by PC and the PC1 market proxy correlations are provided in Appendix A.

4.4 Portfolio Comparison Against SPY

The PCA and Fama-French analyses above show that a small number of latent factors capture most of the cross-sectional risk structure of the stock universe. A natural follow-up is whether this structural understanding translates into better portfolio performance. We construct mean-variance tangency portfolios from the PCA factor covariance and from the Fama-French three- and five-factor covariances, and compare them out-of-sample against SPY and an equal-weight benchmark.

The full-sample PCA tangency portfolio is highly concentrated (Figure 4): the 25% per-stock cap binds for BA, WFC, and AMZN, and 13 of 20 stocks receive zero weight. This reflects the well-known instability of mean-variance optimization, in which small estimation errors in expected returns and covariances produce large swings in weights.

A walk-forward backtest that re-estimates the PCA, FF3, and FF5 tangency portfolios annually using a five-year training window shows the same instability holds out-of-sample. The equal-weight portfolio achieves the highest Sharpe ratio, followed by SPY, while the PCA, FF3, and FF5 tangency portfolios all underperform (Figure 5). PCA tangency also suffers the deepest drawdown (-49.5%) versus SPY's -26.2% (Appendix A).

PCA and Fama-French models are valuable for understanding the structure of risk, but the same factor structure does not automatically produce a superior strategy. The optimization step injects estimation error that outweighs the benefit of factor-aware covariance modeling, and equal-weighting sidesteps the problem entirely.

4.5 Subperiod Stability and the Role of NVIDIA

A full-sample analysis can mask structural change. Because 2023–2025 coincides with the rapid rise of AI-related technology stocks, particularly NVIDIA, we split the sample into 2016–2022 and 2023–2025 to test whether the portfolio's risk-return profile and factor structure remained stable.

Table 2 reports equal-weight portfolio performance across the two subperiods. Although cumulative return is similar across windows (1.029 vs 1.192), the annualized return rises from 12.7% to 27.6% while annualized volatility falls from 22.4% to 16.2%. The Sharpe ratio triples, from 0.57 to 1.70, and maximum drawdown improves from -41.2% to -20.9%. The recent period therefore delivered substantially stronger risk-adjusted performance.

Table 2: Portfolio Performance Comparison Across Subperiods

Metric	2016–2022	2023–2025
Cumulative Return	1.0295	1.1924
Annual Return	0.1267	0.2763
Annual Volatility	0.2241	0.1623
Sharpe Ratio	0.5652	1.7022
Max Drawdown	−0.4118	−0.2095

To examine whether this improvement reflects broad-based gains or concentration in a single name, Table 3 isolates NVIDIA's contribution and PCA loadings across subperiods. NVIDIA's annualized return more than doubled, from 41.7% to 85.4%, while its volatility was essentially unchanged at approximately 49.6%. Its average daily contribution to the equal-weight portfolio rose from 0.000083 to 0.000169, indicating a larger per-day role despite a shorter window.

Table 3: NVIDIA Return Contribution and PCA Loading Across Subperiods

Metric	2016–2022	2023–2025
Portfolio Cumulative Return	1.0295	1.1924
NVDA Avg. Daily Contribution	0.000083	0.000169
NVDA Total Contribution	0.1459	0.1274
NVDA Annualized Return	0.4175	0.8537
NVDA Annualized Volatility	0.4956	0.4960
NVDA PC1 Loading	0.2130	0.2304
NVDA PC2 Loading	0.2649	−0.2775
|NVDA PC2 Loading|	0.2649	0.2775

The PCA structure also shifted. NVIDIA's PC1 loading increased from 0.213 to 0.230, meaning it became more aligned with the dominant systematic factor in the recent period. Notably, PC1 itself explains less variance in the recent subperiod (31.4% vs 45.9%), suggesting the market-wide factor weakened as idiosyncratic AI-driven performance became more prominent. NVIDIA's absolute PC2 loading rose slightly from 0.265 to 0.278; the sign flip is not economically meaningful, since principal component signs are identified only up to $\pm$ and depend on the estimation sample.

Taken together, the subperiod results suggest the portfolio became both more profitable and more concentrated on NVIDIA's growth-driven risk in 2023–2025. The growth-versus-value reading of PC2 is increasingly an NVIDIA story in the recent sample — a caveat worth keeping in mind when extrapolating the factor structure forward.

5 Conclusion

This project shows that PCA is useful for reducing the dimension of stock return data and identifying hidden sources of risk. The first principal component captures systematic market risk, PC2 captures growth-versus-value style rotation, and PC3 captures a profitability and investment dimension. These interpretations are supported by both stock loading patterns and Fama-French regression validation.

However, the portfolio comparison shows that statistical sophistication does not guarantee better investment performance. SPY remains a strong passive benchmark, and the equal-weight portfolio outperforms the optimized PCA and Fama-French tangency portfolios in this backtest. Therefore, PCA's main contribution in this project is factor discovery and risk interpretation rather than guaranteed market outperformance.

The subperiod analysis adds an important caveat. Risk-adjusted performance improved substantially in 2023–2025, but this improvement is closely tied to NVIDIA's stronger loading on the dominant PCA factors. This suggests that the factor structure is not fully stable across regimes and that PCA-based interpretations should be revisited as the market environment evolves rather than treated as static descriptions of risk.

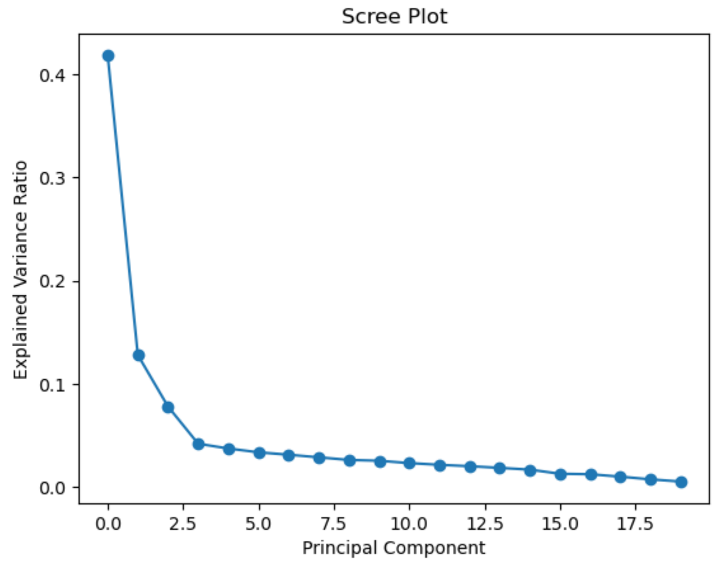

Figure 3: Scree plot of explained variance by principal component.

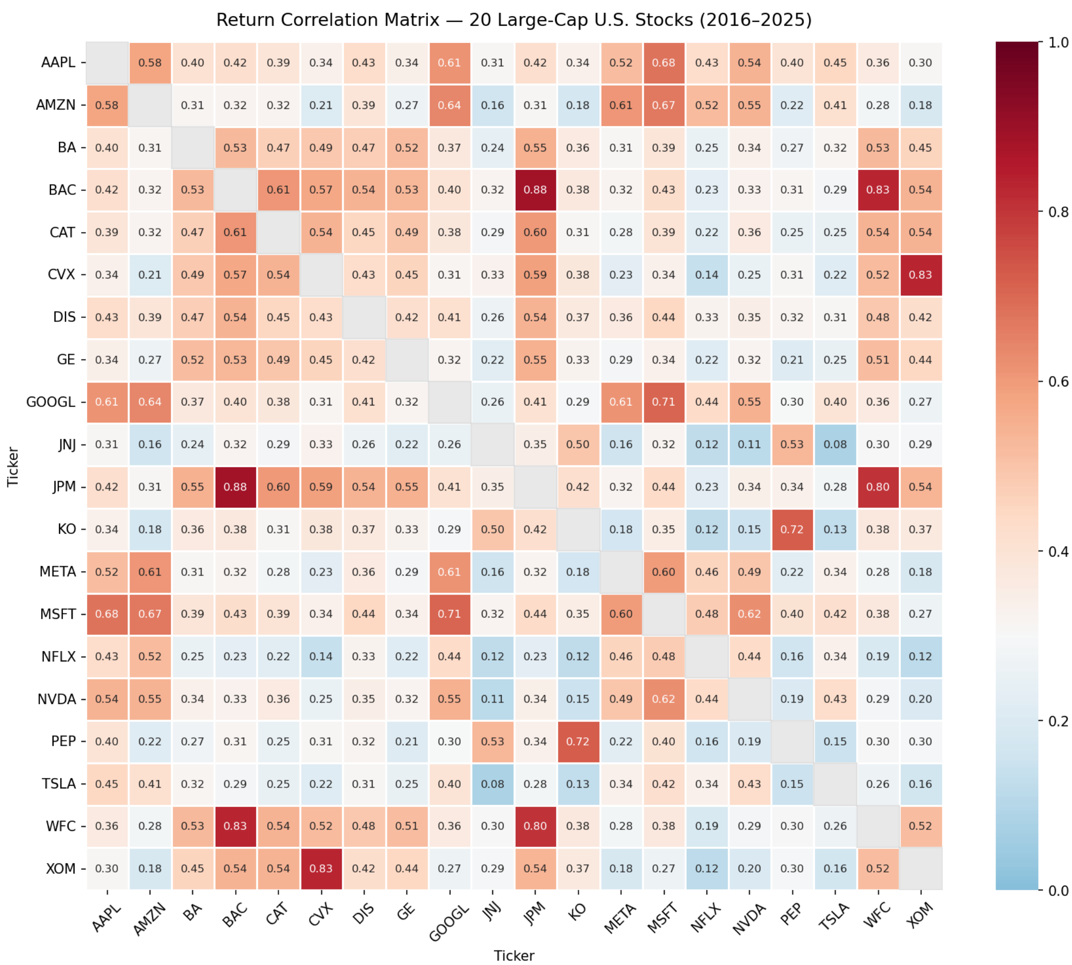

Figure 4: Pairwise return correlation matrix. Within-sector clustering among financials ($\rho > 0.80$) and technology is particularly pronounced.

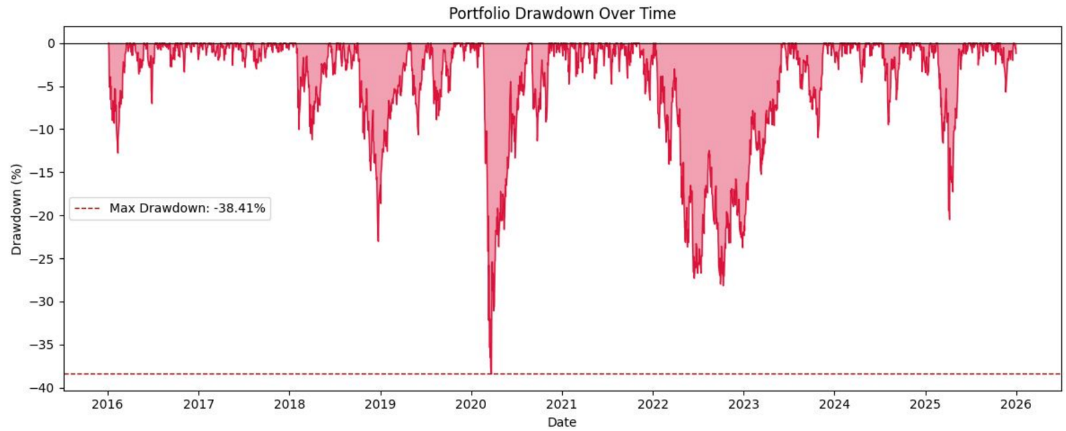

Figure 5: Drawdown of the equal-weight portfolio over time.

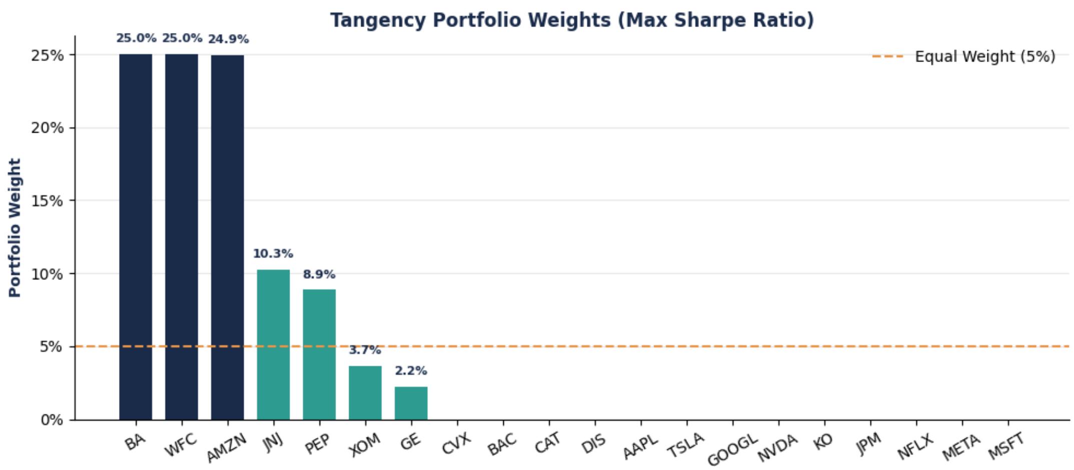

Figure 6: Tangency portfolio weights from PCA-based mean-variance optimization with a 25% per-stock cap.

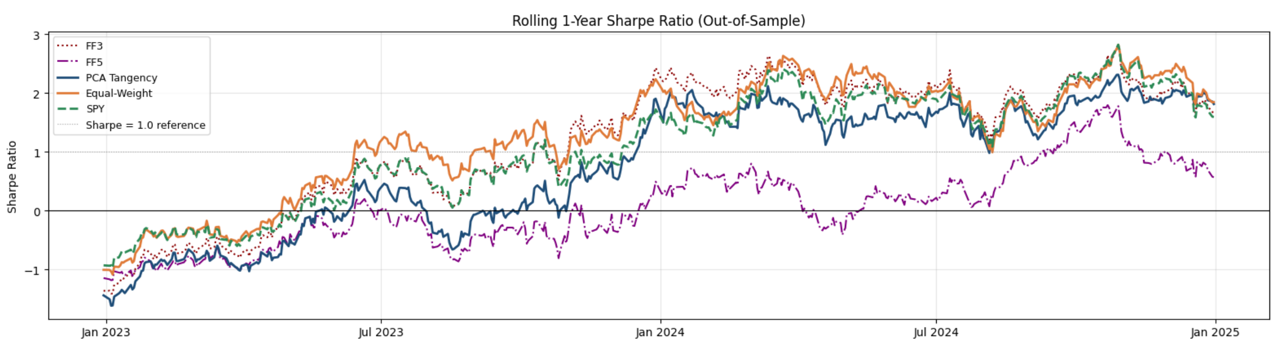

Figure 7: Rolling one-year out-of-sample Sharpe ratios.

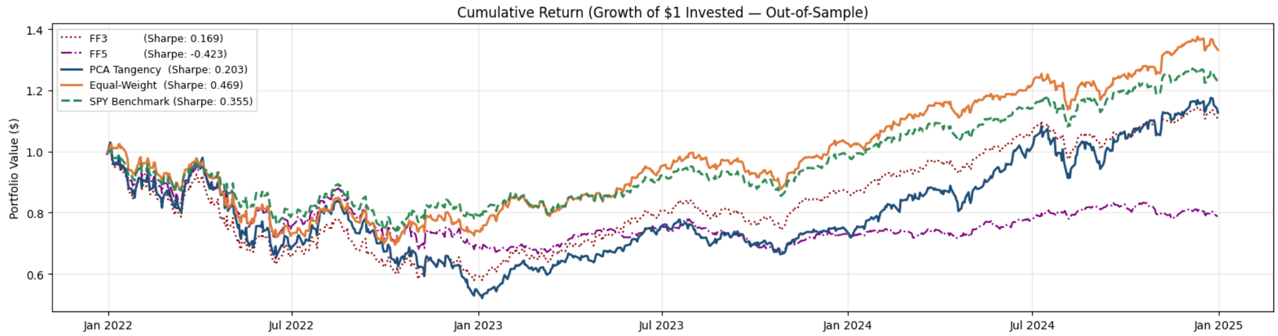

Figure 8: Out-of-sample cumulative return (growth of $1 invested).

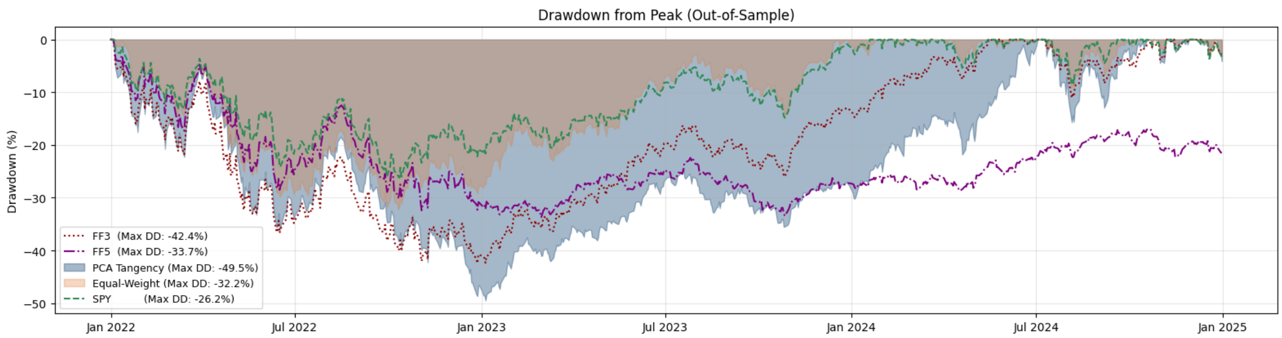

Figure 9: Out-of-sample drawdown from peak.

The efficient frontier compares in-sample risk-return tradeoffs under a 25% position cap. The PCA tangency portfolio achieves the highest in-sample Sharpe ratio of 1.124, above the minimum-variance portfolio at 0.528 and the equal-weight benchmark at 0.706. The PCA tangency portfolio is highly concentrated, with BA receiving one of the largest weights and the 25% cap binding for BA, WFC, and AMZN.

Table 4: Efficient frontier portfolio statistics.

Portfolio	Annual Return	Annual Volatility	Sharpe Ratio
MVP (PCA Frontier)	0.1034	0.1540	0.528
Tangency (PCA Frontier)	0.2817	0.2311	1.124
Equal-Weight Benchmark	0.1714	0.2116	0.706

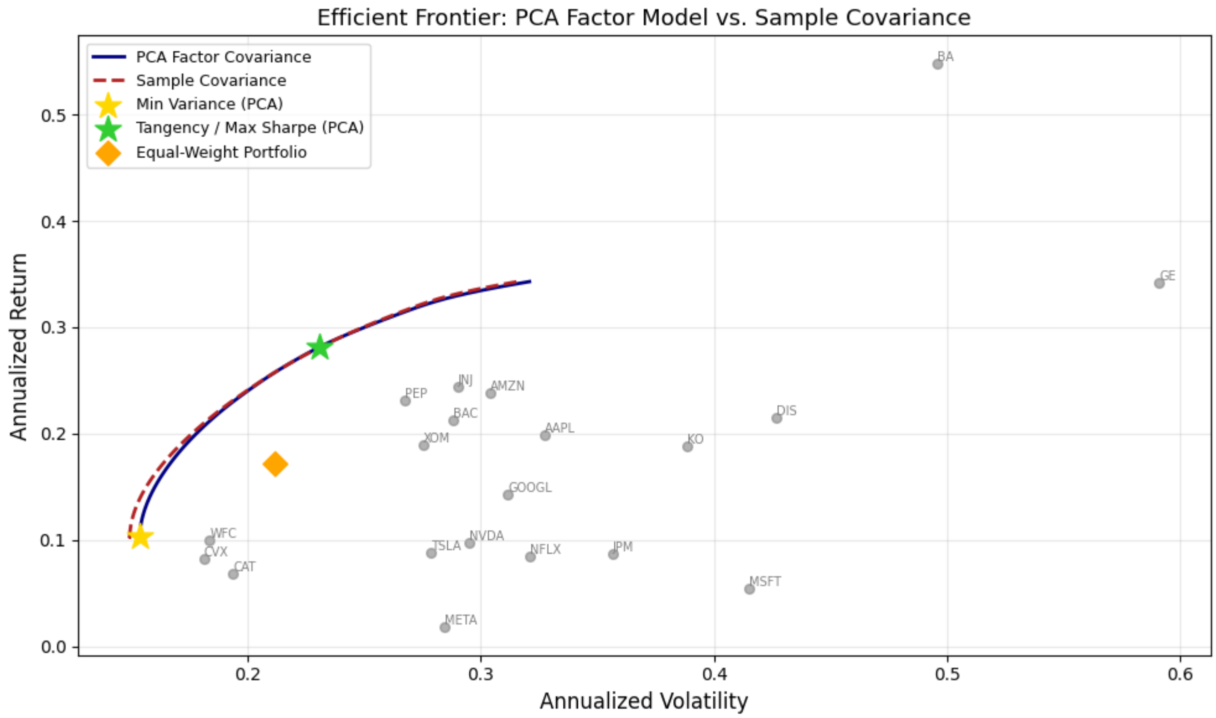

Figure 10: Efficient frontier comparison between the PCA factor covariance model and the raw sample covariance model. The plot highlights the minimum-variance portfolio, tangency portfolio, equal-weight benchmark, and individual stock positions.